

OXG Space

Building Edge Cell Towers in Space

Senior Ka-Band Phased Array PCB Design Engineer

Ka-Band Phased-Array Antenna Electronics

Team	Location	Employment Type	Seniority
RF and Antenna Systems	Bengaluru, India	Full-time	Senior, Staff, or Principal depending on experience

About OXG Space

OXG Space is developing advanced inter-satellite communications and edge-computing systems that combine electronically steered phased arrays, high-throughput communication links, and onboard processing. We are building FPGA-based signal-processing platforms for digital and hybrid beamforming, modem implementation, and real-time communication and AI workloads.

The Role

We are seeking a Senior Ka-Band Phased Array PCB Design Engineer to lead the design and development of high-performance phased-array antenna electronics for satellite and advanced wireless communication systems.

You will be responsible for the complete PCB development lifecycle — from RF architecture and stack-up definition through schematic capture, layout supervision, fabrication, assembly, bring-up, and RF validation. The role requires deep expertise in Ka-band RF design, phased-array architectures, antenna-in-package or antenna-on-PCB implementations, and high-density mixed-signal PCB development.

You will work closely with antenna, RFIC, mechanical, thermal, digital, FPGA, manufacturing, and systems engineering teams to deliver flight-ready phased-array hardware.

Key Responsibilities

- Lead the design of Ka-band phased-array antenna PCBs operating across applicable uplink and downlink frequency bands.
- Develop PCB architectures integrating antenna elements, beamformer ICs, RF front-end components, frequency-conversion circuitry, power distribution, control electronics, and high-speed digital interfaces.
- Design antenna-on-PCB, antenna-in-package, tile-based, or modular phased-array solutions.
- Define low-loss, high-layer-count RF PCB stack-ups, materials, copper structures, surface finishes, impedance requirements, and fabrication tolerances.
- Perform transmission-line, transition, via, connector, filter, matching-network, and RF power-

distribution design at Ka-band frequencies.

- Design and optimize microstrip, stripline, grounded coplanar waveguide, substrate-integrated waveguide, and other RF interconnect structures.
- Develop low-loss transitions between PCB structures, beamformer ICs, RF connectors, waveguides, antennas, and packaged devices.
- Manage RF routing, phase matching, amplitude matching, channel-to-channel isolation, polarization purity, and array calibration requirements.
- Address coupling, scan blindness, grating lobes, surface-wave effects, thermal deformation, and manufacturing variation within the array design.
- Collaborate with antenna engineers to optimize element spacing, array geometry, feed architecture, polarization, scan range, gain, sidelobe performance, and axial ratio.
- Perform or support electromagnetic simulation using tools such as Ansys HFSS, CST Studio Suite, Keysight ADS, or equivalent.
- Conduct signal-integrity and power-integrity analysis for high-speed digital control, clocking, memory, and data-converter interfaces.
- Design robust power-distribution networks for beamformer ICs, power amplifiers, low-noise amplifiers, converters, FPGAs, and supporting electronics.
- Work with mechanical and thermal teams on heat spreading, cold-plate interfaces, shielding, enclosure integration, flatness, and coefficient-of-thermal-expansion constraints.
- Select RF laminates, prepregs, adhesives, connectors, shielding materials, and packaging technologies appropriate for Ka-band and space environments.
- Create schematics, PCB layout constraints, fabrication drawings, assembly drawings, impedance tables, material specifications, and manufacturing notes.
- Review PCB layouts and provide detailed guidance to internal or external PCB layout engineers.
- Work directly with PCB fabricators and assembly partners to resolve design-for-manufacturing, yield, registration, plating, via, lamination, and assembly challenges.
- Develop RF test coupons and calibration structures for validating fabricated board performance.
- Lead board bring-up, troubleshooting, and characterization using vector network analyzers, spectrum analyzers, signal generators, noise-figure analyzers, oscilloscopes, and near-field or far-field antenna test systems.
- Correlate simulated and measured results and implement corrective design changes.
- Support phased-array calibration, beam steering, beam pattern measurements, gain-to-noise-temperature measurements, EIRP verification, and over-the-air testing.
- Perform design reviews, failure analysis, root-cause investigations, and design-risk assessments.
- Support environmental qualification, including thermal cycling, thermal vacuum, vibration, shock, radiation, and reliability testing.
- Mentor junior engineers and establish internal design rules, simulation methodologies, documenta-

tion standards, and PCB review processes.

Required Qualifications

- Bachelor's or Master's degree in Electrical Engineering, RF Engineering, Microwave Engineering, Applied Physics, or a related discipline.
- At least 7 years of experience designing RF, microwave, or millimeter-wave PCB hardware.
- Demonstrated experience developing hardware operating at Ka-band or comparable millimeter-wave frequencies.
- Strong understanding of phased-array antenna architectures, beamforming, RF channel matching, array calibration, and beam-steering principles.
- Experience designing multilayer RF PCBs using low-loss materials such as Rogers, Taconic, Panasonic Megtron, Isola, or equivalent laminates.
- Strong knowledge of PCB transmission lines, impedance control, dielectric loss, conductor loss, dispersion, surface roughness, via parasitics, resonances, and electromagnetic coupling.
- Experience integrating RFICs, beamformer ICs, LNAs, PAs, mixers, PLLs, frequency synthesizers, filters, switches, attenuators, and phase shifters.
- Proficiency with at least one electromagnetic simulation platform.
- Proficiency with professional PCB design tools such as Cadence Allegro, Siemens Xpedition, Altium Designer, or equivalent.
- Experience working with PCB fabrication and assembly vendors on complex, high-frequency designs.
- Hands-on experience with RF laboratory equipment and millimeter-wave measurement techniques.
- Ability to independently lead complex PCB designs from requirements through production and qualification.
- Strong technical documentation, design-review, troubleshooting, and cross-functional communication skills.

Preferred Qualifications

- Experience designing active electronically scanned arrays for satellite communications, radar, 5G, defense, aerospace, or high-capacity wireless systems.
- Experience with Ka-band satellite systems and communication links.
- Experience with dual-polarized arrays and advanced polarization techniques.
- Experience integrating phased arrays with FPGAs, RFSOCs, and high-speed digital systems.
- Familiarity with hybrid beamforming and tile-based architectures.
- Experience with advanced PCB technologies including HDI and microvias.
- Understanding of thermal management for high-density RF systems.
- Experience designing space-qualified electronics.

- Familiarity with aerospace PCB standards.
- Experience transitioning prototypes to production.
- Familiarity with RF calibration and testing.
- Experience leading technical teams.

Success in This Role

Within the first year, the successful candidate will:

- Establish PCB architecture and stack-up for a Ka-band phased-array antenna tile.
- Complete prototype hardware development and validation.
- Demonstrate strong correlation between simulation and measurement.
- Resolve key RF and manufacturability challenges.
- Establish reusable design rules and processes.
- Support successful beam-steering demonstrations.
- Develop a path to scalable production.

Role Details

Field	Details
Work Arrangement	On-site or Hybrid
Travel	Occasional